

Strategy Design Overview

Before using AI Studio, I did some offline deep research (with ChatGPT as I've found this the best deep research AI) on what is realistically extractable from the QQQ dataset and what kinds of regime/alternative-data ideas might make sense. I then used that as scaffolding so that my prompts in Gemini started from concrete hypotheses instead of just asking it to maximize Sharpe.

The process begun with optimising the model performance from technical indicators derived from QQQ and other regular datasets, VIX, 10-2Y Treasury Yields, correlation between S&P500 stocks etc. and then moving onto alternative data optimisation to ensure I get the best out of both. At the beginning of my Gemini thread, my prompts were deliberately exploratory. I asked Gemini to load the QQQ data, build a reasonable library of technical/volatility features, apply the 2000–2015 train / 2016–2021 validation split, and backtest different signal families. I also told the model to try 30+ different trading persona such as, volatility, buy the drop etc., and then prompted to combine them together by testing multiple different combinations and rules.

Originally, I started by using a regime switching model (Gaussian Mixture Model mainly) to have fixed leverages during different regimes, but this did not perform well – didn't pick up the discrete states well. I then moved onto training an ML model that takes in multiple orthogonal signals and outputs a probability of 0-100 (where 0 is 0% probability of QQQ going up the next day and 100 is 100% probability of QQQ increasing the next day). Random Forest models performed best here as other models such as, gradient boosting and decision trees, overfit, underperformed or both.

From here, I would prompt gemini to go through my script and understand where it was underperforming and what indicators are causing this. Then I prompted again to research 30+ different signals that we can extract to fill the underperforming periods, and then I sourced the datasets. After this, gemini would run a detailed correlation matrix, precision, information coefficient, and accuracy scores and we would pick together what signals are uncorrelated with our current set of features and have predictive power. Overall, I generated 50+ signals that I then ran a detailed grid search over, where I helped the model by defining 15-20 "must-have" signals that I knew were very uncorrelated and had large predictive performance over the QQQ. I also ran a grid search over many different leverage strategies for every random forest model trained here. Next, I noticed that changing seeds did have quite large variance and Evan informed I could likely add some more OHLCV features. I sourced those extra features with the help of gemini and retrained the model. This time however, I tested 5 random seeds and if there was a difference greater than +- 10% over different seeds the model was thrown out. This yielded a model that was not overfit and was well optimized.

Next, I added the TLT into a combined QQQ and TLT strategy. I would have liked to use DBMF originally as it more uncorrelated with the QQQ then TLT but this doesn't have data dating back 2019 so I stuck with TLT (which is still uncorrelated). I used the same feature set that I used for the QQQ dataset and also prompted gemini to research indicators that I don't have that are informative for the TLT. Then I again added the uncorrelated signals that have

high predictive power into the feature set and ran a similar grid search as the QQQ – with the seed variance filter too. I changed the “must have” features here as well as changed `max_features` to None (rather than default `sqrt`) for the random forest models. This performed much better and also used less features, this is likely because TLT is very well described from just a few macro and other features here.

In combining the QQQ and TLT strategy I tried Kelly-weighting, Markowitz portfolio theory/min-variance, inverse volatility filter, and a ML meta-learner. The ML meta-learner performed best here via a similar grid search as QQQ and TLT with added “must-have” features of the probabilities outputted by the QQQ and TLT ML models and multiple different rolling-window simple correlation and variance formulas. I also noticed that the model outputted weights largely in the range 0.4-0.6, therefore, as QQQ is higher variance and will be our main money maker here I inputted override weights, where if the probability from the QQQ ML model exceeded both low (high short probability) and high (high long probability) thresholds we overweighted the QQQ weight that the meta-learner outputted. I also inputted an override that if volatility for QQQ was very high we underweighted QQQ (this didn’t change much though and really only worked for one edge case).

Throughout the whole process, I was sure to only pick the best model and leverage settings based on the validation and train Sharpes and used the out-of-sample blind Sharpe only for pure out-of-sample post-mortem analysis.

Gemini Prompts:

User

I am in the final round of Quanta Ventures Fellowship which requires a take home task that uses Google's AIStudio to generate a trading strategy that generates the highest Sharpe possible, specifically ≥ 2.0 and performs well over the blind sample as well as the validation sample. We have train dates from 1/1/2000 through 12/31/2015 and validation dates from 1/1/2016 through 12/31/2021, there should be no lookahead bias in generating the signals and we should try to use 30+ individual signals to generate the trading strategy, we can only generate trades on the QQQ with a leverage between -1 and +1.5. First step that you should do is read the Quanta Fellowship Finalist Project that I've attached to this prompt to understand the task in depth. Then you should use the links to the papers that I am going to attach as well as the summary in the documents and any other further research that you view as necessary to generate the first iteration of our trading strategy based on all this information. For this time I will not upload alternative data as this is the first test, you should evaluate multiple different strategies that you are able to come up with through the research that I have already conducted and any further research you conduct. Then we will iterate by adding alternative data and looking at other signals etc. to get the best result. We should only be touching the train and validation set right now, we will not iterate our strategy over the blind tests, that will be for later. You should plot a graph with the equity curves of the multiple strategies that you use and report their Sharpe Ratios over both the train and test period, if at any stage of this initial run through you need any more information to do your best of the initial run stop doing the run, let me know then I'll answer and you'll continue from where you left off. You should use the Train & Validate.csv that I've attached in this prompt to run the strategies and at the end there should be a graph plotting the equity curves of

all the strategies and a return of the Sharpe ratio that each gives. You should test multiple different strategies and ideas, the more the better, and you should test multiple different ways of combining strategies into an ensemble, ensure that you show me the equity curve of the proposed strategies, you should be running multiple different trading rules and ways of combining them, think 10+ at least. We ran this prompt once and you tried to literally parse the whole csv by doing `csv = "...."`, just do it normally with pandas, i.e. `pd.read_csv("...")` this is much more efficient. For this first step you should generate 15+ different trading ideas/personas with each based on a lot of indicators and well researched/well known trading personas. Then show me the equity curves and the sharpes of all these ideas then we will look into different methods to combine them in a separate prompt, then we will look at alternative datasources. Here is the research I have done:

Note I didn't include the research here in this prompt demonstration as it is very long (gemini had to take it as text not a pdf, as when it is pdf it turned off code execution).

User

Alright nice work there, now lets delve into ways that we can use alternative indicators to first of all help us not generate the losses that still occur during the 2018-2019 and 2022-2023.5. To do this you should understand what is triggering us to hit these losses (i.e. we are in the wrong state and therefore generating longs rather than shorts). This will involve you analysing the losses that occur and what features are causing the model to output such a probability. Report back to me with specific reasons why we are generating losses over these periods. Ensure you run the code you write.

User

Alright now in the dataset that I uploaded I have given you new data that I havent used yet, I want you to go research 30+ ways to generate new signals from this and then create the signals and test their correlations to the current strategy that just run which returned a sharpe of 1.64 validation. You should then return to me this correlation matrix - one thing I definetly want you to try is backing out : If you want to know roughly what an "At-The-Money" option cost in 2020, you don't need the specific contract. You can estimate it using the Black-Scholes model if you have:

The QQQ price for that day.

The $\wedge$ VXN (Implied Volatility) for that day.. ensure you ran code you write

User

Okay now, I need to push the strategy to sharpe >2 in the out-of-sample period. We perform great during the train period but during the validation period we drop down a bit and during the blind OOS we drop down alot. we drop down a. lot due to significant drops, indicating that the model is picking the wrong side of the trade multiple times during this period. Go through the results for the blind OOS, understand why our model is making the wrong trades and where, and then understand what in our feature set - the features that you just ran the correlation analysis etc. on - is causing this and therefore what gaps in our feature set we have is. Ensure you run any code that you write and run the analysis on the top 2 strategies we have currently

User

Okay that analysis is good, adding that Sig_RIS_2 feature will not improve us that much. I want you to research 30+ different sources of alternative signals, I am talking about sourcing alternative data to what we already have - we've basically exhausted all feature sets from this data here -, that will fix our issues. Some general thoughts to guide your research, indicators like Z(VIX)-Z(VVIX), SMA_5(CPCE) etc. could work.

User

Alright now I have uploaded a csv with multiple new columns that you asked for, VVIX, VXN, DF, VIX3M and some others. Use your research and create those features that you identified (some you wont be able to do as I cant source the data). From this, run me a correlation matrix of these features with the features that we used in the previous strategy (the ones we ran the IC on). Return to me the correlation matrix as well as IC for new variables for both train, and test (valid) period. Ensure you run any code that you write, and ensure you use the correct input file, it should be the input file that matches to 'master_df_2.csv'

User

Okay that is great, now we are going to add these features into our feature set, each iteration should have all the other signals from the random forest and gradient boosting respectively (as we are running this on both) in it as well as a mix of these other new signals, given the correlations and your understanding of the features and there accuracy results etc. you should be able to understand which ones will compliment each other and are generally uncorrelated with eachother, obviously if a signal is uncorrelated with the strategy but correlated with another signal that is already in the strategy or outside the strategy there is no need to add both signals into the strategy. You should test 30+ different combinations of signals and report back to me the top 5 best performing strategies. Again you are only to train on the training period and we should ensure good performance on both train and validation sets, here you should run the analysis on both random forest and gradient boosting. Ensure you return to me the equity curve for the top 5 strategies as well as the sharpe ratios for train and validation. ensure you run any code you write . I am going to help you out at the start here, you are running into an issue where it says that you cannot find the QQQ data file, so ensure you set up that correctly, it is still given in the first prompt

User

Now given we have this large set of signals, i want you to research 30+ ways of trading e.g. momentum, volatility filtering, bull/bear etc. And then look at my feature set and report back to me which ones of these popular trading strategies do we not have indicators for

User

Okay I have been giving the task to generate a strategy that gets a sharpe >2.0. I have developed a strategy that generates a pool of 45+ uncorrelated signals using macro data and then trains a small decision tree to output a probability from 0 to 100 of the QQQ going up, and then based on that output we generate leverage from -1 to 1.5. Now my task is this: If you think you've plateaued on QQQ then you're welcome to jump to Phase 2 which can have a huge boost

which is you can add another asset in the mix. Essentially do a very similar process on a totally different, less correlated asset (TLT, GLD, DBMF for example) and then combine a strategy on that one as a sleeve in addition to QQQ. That's where real magic happens. I want you to act as a senior quantitative researcher throughout the process together as you are tasked with generating my set of uncorrelated signals. here are the constraints of the task to keep throughout: TRAIN:

You must train your signals using only data from 1/1/2000 through 12/31/2015

VALIDATE: You must test your signals on the validation test period of 1/1/2016 through 12/31/2021

You shouldn't develop new signals on validation 1/1/2016 through 12/31/2021, but you can tune your signals and use it for model selection and looking at combined signal performance.

The Blind Holdout Period is 1/1/2022 - 6/30/2025. Never iterate using this period.

You should only look at the Blind Holdout Out of Sample performance after you've completed signal development and your overall model. *Never* iterate using the blind out-of-sample period. If you see that your model doesn't perform well enough in the blind out of sample, do a post mortem with the help of Gemini to understand why, then go back to the training period and try to improve the process.

You can use leverage from -1 (100% short) to +1.5X (150% Long) per day. Each trade/change of leverage must happen at the close and then be held until the next close. No stop losses can be used.

Robustness (Sensitivity Testing): You must demonstrate that your signals are not overfit. A $\pm 10\%$ shift in your key parameters (e.g., moving average lengths, RSI thresholds, volatility windows) should not result in a strategy collapse. If changing a 20-day lookback to an 18-day lookback destroys the alpha, the signal is rejected.

You must ensure there's no lookahead bias. All data and signals must be calculated on information known at the time of the trade or before. For example, it would be invalid to create indicators or signals from the entire dataset and then use those indicators to make trading decisions earlier on (like Z-scores, or averages over the whole period, etc). You can ask Gemini to test this for you at a signal or portfolio level by analyzing the code. First of all, I want you to research 30+ different assets and return to me the top most likely to be uncorrelated with QQQ, the idea here is that we will then trade both simultaneously in one portfolio.

User

Okay, I have two strategies combined as you can see here in the attached script, I want to test some more advanced ways of combining them such as, markowitz portfolio theory, training a small ML model to spit out weightings, etc. I want you to research 15+ best methods of combining two uncorrelated asset. heres the current combination script

User

I have a meta learner that takes in QQQ and TLT probabilities of being long that is outputted by a random forest from 0-1 then I generate leverages. Then the meta learner takes in the probability and other features and then outputs a weighting from -1 to 1.5. I want you to research the top 15+ signals that a meta learner aimed at producing portfolio weights needs. Report back to me the top ones, here is the current script, analyse it first and see what is missing from the meta-learner features based on your research and report back to me:

User

Okay go through this script and determine if there is any lookahead bias, be very strict here we need no lookahead bias at all, this includes calculating z-scores shifting macro data, weights we have produced etc.: